

4.4.4 How Strong is your password?

Description

Now that you've had time to play around with the password strength tester, write down your answers to the following questions. **For 2 through 5, include how long it would take to guess the password.**

1. Why do you think websites commonly require you to have at least 8 characters, and a mix of uppercase characters, lowercase characters, numbers, and symbols?
2. Try making 8 character passwords using only lowercase alphabet characters (a-z). What is the strongest password you can make?
3. Now try mixing upper and lowercase characters, still only 8 characters total. Now, what is the strongest password you can make?
4. Now try adding numbers and symbols, still only 8 characters total. Now, what is the strongest password you can make?
5. Now, using only lowercase characters, make a password that is 20 characters or longer. Is this password better or worse than an 8 character password with letters, numbers, and symbols?
6. As you play around with different types of passwords, what seems to be the single factor that guarantees a strong password?

If you would like to use the password checker to help answer questions, go to <https://www.comparitech.com/privacy-security-tools/password-strength-test>.

Solution Code

default.txt

Student answers will vary.

1. Why do you think websites commonly require you to have at least 8 characters, and a mix of uppercase characters, lowercase characters, numbers, and symbols?
The more complicated a password is, the harder it will be to guess. By mixing lower and uppercase letters, symbols, and numbers, the possible combinations that a guess-and-check attack would need to guess is
2. Try making 8 character passwords using only lowercase alphabet characters (a-z). What is the strongest password you can make?
Sample response: qmpwyxzk would take 4 months to guess
3. Now try mixing upper and lowercase characters, still only 8 characters total. Now, what is the strongest password you can make?
Sample response: QmPWYxzK would take 43 centuries to guess
4. Now try adding numbers and symbols, still only 8 characters total. Now, what is the strongest password you can make?
Sample response: Q}7~Y6zK would take 1000 years to guess
5. Now, using only lowercase characters, make a password that is 20 characters or longer. Is this password better or worse than an 8 character password with letters, numbers, and symbols?
Sample response: qmpwyxzkqmpwyxzkxpzl would take 3 thousand trillion years to guess
6. As you play around with different types of passwords, what seems to be the single factor that guarantees a strong password?
Very long passwords seems to be the best factor. Even if every character is of the same type (numbers, letters, etc), having 20 characters to guess takes an incredibly long time. If you had 26 characters to